

DATA STRUCTURES AND ALGORITHMS – 3**Y25 - 2026-2027****Project Abstract Submission Form**

Sec No- 1	Team No- 23	Project Title- Statistical Analytics Dashboard Using Parallel Algorithms
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1. Abstract

The Statistical Analytics Dashboard is a software application designed to efficiently process and analyze large numerical datasets using parallel computing techniques. The system computes important descriptive statistics such as mean, variance, minimum, maximum, cumulative sums, and frequency distributions. To improve computational efficiency, the project implements Parallel Prefix Sum and Parallel Reduction algorithms and compares their performance with sequential approaches. The dashboard presents the computed statistical results along with graphical reports that clearly visualize execution time and performance differences between sequential and parallel processing. The main objective of this project is to demonstrate how parallel algorithms can accelerate the analysis of large-scale numerical data while maintaining accurate statistical results. Overall, the project provides an interactive and easy-to-understand platform for statistical analysis and performance evaluation, highlighting the practical advantages of parallel processing in data-intensive applications.

2. Team Details

Roll Number(s)	Name of the Student
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3.Problem Statement:

Processing large numerical datasets sequentially can require significant execution time, especially when multiple statistical operations must be performed. The problem is to develop a statistical analytics dashboard that efficiently computes descriptive statistics over large datasets and evaluates the performance benefits of parallel processing. The system should calculate mean, variance, minimum, maximum, cumulative sums, and frequency distributions using appropriate sequential and parallel algorithms. It should specifically use Parallel Prefix Sum for cumulative computations and Parallel Reduction for operations such as sum, minimum, and maximum. The system must compare sequential and parallel execution times and present the results through graphical reports.

4.Objective:

The main objective of this project is to develop a statistical analytics dashboard that efficiently analyzes large numerical datasets using parallel algorithms. The system aims to compute mean, variance, minimum, maximum, cumulative sums, and frequency distributions accurately. It also aims to demonstrate the practical use of Parallel Prefix Sum and Parallel Reduction algorithms by comparing their execution performance with sequential methods. The project further provides graphical performance reports to make the comparison clear and understandable.

5.Frameworks / Platforms / Software Used

Programming Language: Java

Algorithms: Parallel Prefix Sum and Parallel Reduction

Concepts: Descriptive Statistics and Parallel Computing

Dashboard / Visualization: Graphical Performance Reports

IDE: IntelliJ IDEA / Eclipse / Visual Studio Code

Version Control: Git and GitHub